

Advanced Neural Approaches for Extractive and Abstractive Text Summarization in Health Information Retrieval

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Abstract

Summarization of medical texts is a difficult but clinically significant task. This paper addresses two tasks from the Medical Questions and Answers (MEDIQA) 2021 - Consumer Health Question (CHQ) summarization, and Multi-Answer Summarization (MAS). For part 1, we fine-tune and evaluate two sequence-to-sequence models - FLAN-T5-base and SciFive-base on the MeQSum dataset [2], with a Lead-1 baseline. SciFive scores ROUGE-L of 0.507 and BERTScore F1 of 0.936, outperforming FLAN-T5 by +0.177 and +0.040 respectively. For part 2, we introduce a new three-step approach of Longformer-based extractive sentence with scispaCy biomedical NER-based re-ranking classification, scispaCy NER-guided re-ranking and abstractive fusion. We also perform a cross-domain evaluation by evaluating 500 examples from PubMed for generalisation, and did a experiment with fusion, by fine-tuning SciFive and observed that the fusion model outperforms FLAN-T5. The SciFive fusion scores ROUGE-1 +0.116 higher than FLAN-T5 on the MEDIQA test set, but there are challenges with domain gap and limited data. We present detailed error analysis, limitations and propose actionable future directions.

1. Introduction

As health information has grown on the web, there is a pressing need for automated methods to summarise complex medical information into accurate summaries. Consumers' questions about health are typically long, charged and poorly articulated, making it hard for clinical information systems to display relevant responses. At the same time, systems that aggregate multiple source documents need to merge potentially overlapping information into a single response.

The MEDIQA 2021 shared task [1] identifies three challenges. Our research focuses on the first two: CHQ summa-

rization (Task 1) and Multi-Answer Summarization (Task 2). Task 1 is a task of condensing a long health question into a reformulated question. Task 2 requires generating a consolidated summary of several ranked answer documents.

Models pre-trained using encoder-decoder architecture, such as T5 [3] and BART [8], have performed well on general summarization tasks. However, extending them to the informal language of consumer health text while maintaining accuracy is not straightforward. Biomedical pre-training variants [5, 12] partially solve this problem, but the improvement for consumer-facing tasks has not been systematically studied.

This paper makes the following contributions:

- A comprehensive study comparing general (FLAN-T5) and (SciFive) fine-tuned models on CHQ summarization, with error analysis across five error types.
- A new multi-answer summarization pipeline with three stages Longformer extractive scoring [6], scispaCy NER re-ranking with answer-quality weighting [7], and FLAN-T5 abstractive fusion.
- A zero-shot cross-domain test on PubMed scientific abstracts, showing domain shift patterns and register collapse under zero-shot conditions.
- A multi-domain fusion experiment using MEDIQA and PubMed data to train SciFive as the Stage 3 abstractive model.

2. Related Work

2.1. Abstractive Summarization

Abstractive summarisation was introduced with the encoder-decoder paradigm by BART [8] and T5 [3]. PEGASUS [9] introduced gap-sentence generation as a pre-training objective specifically designed for summarisation, achieving state-of-the-art results on a number of benchmarks. For both tasks, we adopt the paradigm of large-scale pretraining and task-specific fine-tuning of these models.

2.2. Biomedical Text Summarization

BioBART [12] and SciFive [5] further pre-train on biomedical corpora such as PubMed abstracts and PMC full texts to improve performance on downstream clinical NLP tasks. MeQSum, a large-scale dataset of consumer health questions and expert-written summaries, was introduced by Ben Abacha and Demner-Fushman [2]. Their original system was based on attention-based neural networks; we follow up on this line of work with instruction-tuned and biomedical transformer models.

2.3. Extractive Summarization

Longformer [6] extended BERT to 4,096-token contexts with sliding-window attention and thus is applicable for long-document sentence classification. SummaRunner [13] proposed a hierarchical recurrent model for extractive sentence scoring. Our Stage 1 uses Longformer as a binary sentence classifier trained on gold extractive summaries.

2.4. Multi-Document Summarization

PRIMERA [14] and hierarchical approaches [15] perform multi-document fusion by jointly encoding multiple sources. Answer aggregation in the biomedical domain also needs to handle topic overlap, redundancy and variation in answer quality. We tackle these problems in a pipeline with answer-quality weighting and NER-guided consensus boosting.

3. Methodology

3.1. Task 1: CHQ Summarization

3.1.1. Dataset

We use the MeQSum dataset [2] comprising 1,000 (CHQ, Summary) pairs, split 800/100/100 (train/validation/test) with seed 42.

3.1.2. Models

We fine-tune two encoder-decoder models under identical hyperparameters (Table 1) for a controlled comparison.

FLAN-T5-base [4] is a 250M parameter instruction-tuned T5 model, trained on a mixture of NLP tasks across domains. For each CHQ input, we prepend the instruction prefix “Summarize this health question into one concise question:”.

SciFive-base-PubMed_PMC [5] is a T5 variant that has been further pre-trained on PubMed and PMC biomedical text, providing domain-specific vocabulary and semantic alignment that general-purpose models lack.

3.1.3. Loss Function and Optimizer

The **AdamW optimizer** [18] is used to train both models. It has a linear learning rate schedule and a warmup period

Table 1. Task 1 Training Hyperparameters

Hyperparameter	Value
Learning rate	5×10^{-5}
Batch size (train/eval)	8 / 16
Epochs	5
Warmup steps	50
Max input / output tokens	512 / 64
Decoding	Beam search ($k=4$)
Best model metric	ROUGE-L (validation)

of 50 steps. The goal of training is to minimise the standard cross-entropy loss over the target token sequence, which is done with teacher forcing. For Task 1, label smoothing is turned off so that token predictions stay sharp. In Stage 1, the Longformer classifier uses binary cross-entropy loss on sentence-level include/exclude labels that come from string-similarity matching against gold extractive summaries.

3.1.4. Task 1 Architecture

Figure 1 illustrates the Task 1 pipeline. The instruction prefix is prepended to the CHQ before tokenization. The encoder produces contextual representations; the decoder generates the concise summary autoregressively.

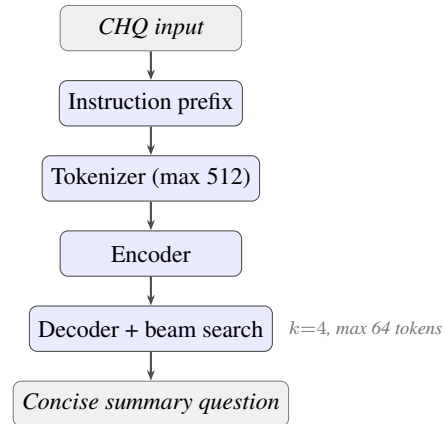


Figure 1. Task 1 pipeline. FLAN-T5 and SciFive evaluated identically.

3.2. Task 2: Multi-Answer Summarization

3.2.1. Dataset

For Task 2 we use two datasets:

The main dataset is a JSON file of 156 consumer health questions from the MEDIQA 2021 shared task [1]. For each question, multiple answer documents from the web are provided, annotated with quality ratings (*1-Bad* to *4-Excellent*), a gold extractive multi-answer summary (*extractive_summary*), and a gold abstractive multi-

answer summary (abstractive_summary). We split this dataset 109/24/23 (train/val/test, seed = 42).

We also use the PubMed summarisation dataset [16] (ccdv/pubmed-summarization) for cross-domain evaluation and multi-domain fusion. This dataset contains full scientific article bodies paired with structured abstracts as gold summaries. We take 500 examples from the test split for zero-shot evaluation, and 500 examples from the training split (450 train / 50 val) for SciFive fusion training. We map PubMed examples to the MEDIQA Task 2 format by taking the article title as the question, splitting the article body into up to six three-sentence chunks as pseudo-answer documents (each rated *4-Excellent*), the full abstract as the abstractive gold, and the first three abstract sentences as the extractive gold proxy.

3.2.2. Three-Stage Pipeline

Figure 2 illustrates the proposed pipeline.

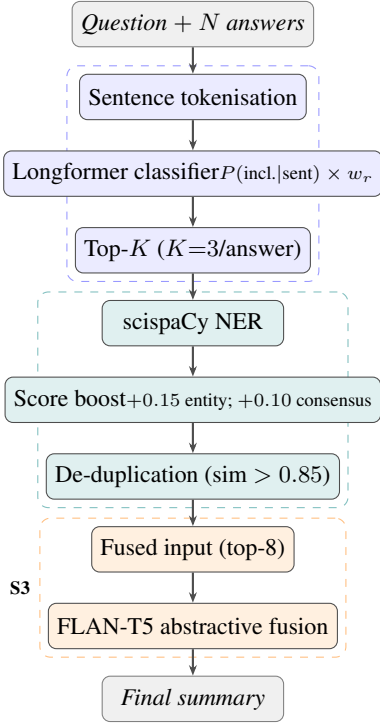


Figure 2. Task 2 three-stage pipeline. S1: Longformer extraction with quality weighting. S2: NER re-ranking. S3: FLAN-T5 fusion.

Stage 1 — Longformer Extractive Classifier. We fine-tune longformer-base-4096 [6] as a binary sentence classifier. Each sentence is scored with $P(\text{include} | \text{sent})$ and multiplied by quality weight $w_r \in \{0.5, 0.75, 0.9, 1.0\}$ derived from the answer rating. Labels are assigned by string-similarity against gold extractive sentences (≥ 0.6 threshold). Top- K ($K=3$) sentences per answer are retained.

Stage 2 — NER-Guided Re-ranking. We apply scispaCy (en_core_sci_sm [7]). The adjusted score is:

$$s'_i = s_i + \delta_{\text{NER}} \cdot \mathbb{I}[\text{ents}(i) \neq \emptyset] + \delta_{\text{cons}} \cdot \mathbb{I}[\text{ents}(i) \cap E_c \neq \emptyset] \quad (1)$$

where $\delta_{\text{NER}}=0.15$, $\delta_{\text{cons}}=0.10$, and E_c is the consensus entity set (entities in ≥ 2 answers). Near-duplicate sentences (similarity > 0.85) are removed.

Stage 3 — FLAN-T5 Abstractive Fusion. Top-8 sentences are concatenated with the question and fed to fine-tuned FLAN-T5-base (max 1,024 input / 200 output tokens, beam search $k=4$).

3.2.3. Cross-Domain Evaluation on PubMed

To assess generalisation, all Task 2 systems are evaluated zero-shot on 500 examples from the PubMed summarisation dataset [16] (ccdv/pubmed-summarization), mapping article body sections to answers and the abstract to the gold summary.

3.2.4. Multi-Domain SciFive Fusion

Motivated by the domain gap in PubMed evaluation, we replace Stage 3’s FLAN-T5 by SciFive fine-tuned on mixed MEDIQA (109 examples) and PubMed (450 examples) with a plain question-plus-sentences format without instruction prefix. Hyperparameters: lr= 2×10^{-5} , 8 epochs, batch size 4, weight decay 0.01

3.3. Implementation Details

All models are made with the Hugging Face transformers library [19] in **PyTorch**. Google Colab with GPU acceleration is used for training and inference. For Task 1, the Seq2SeqTrainer API is used to fine-tune both FLAN-T5 and SciFive with the hyperparameters listed in Table 1. For Task 2, the Longformer classifier uses the Trainer with 2 gradient accumulation steps and a batch size of 4 per device. With Seq2SeqTrainer, predict_with_generate=True, and a generation length of 200 tokens, Stage 3 FLAN-T5 is fine-tuned. The SciFive fusion model has a learning rate of 2×10^{-5} , a weight decay of 0.01, and 8 epochs with a batch size of 4. For reproducibility, all random seeds are set to 42.

4. Evaluation Metrics

4.1. ROUGE

ROUGE [10] measures the n -gram based overlap between the candidate and the reference. **ROUGE- N** :

$$\text{ROUGE-}N = \frac{\sum_{s \in R} \sum_{g_n \in s} \text{Count}_{\text{match}}(g_n)}{\sum_{s \in R} \sum_{g_n \in s} \text{Count}(g_n)} \quad (2)$$

However ROUGE-L uses the Longest Common Subsequence (LCS):

$$\text{ROUGE-L} = \frac{(1 + \beta^2) P_{\text{lcs}} R_{\text{lcs}}}{R_{\text{lcs}} + \beta^2 P_{\text{lcs}}} \quad (3)$$

where $\beta \rightarrow \infty$ weights recall. ROUGE-L captures sentence-level structure without requiring contiguous matches.

4.2. BERTScore

BERTScore [11] computes the semantic similarity using the contextual BERT embeddings. Precision, recall, and F1 are defined as:

$$P_B = \frac{1}{|\hat{x}|} \sum_{\hat{x}_j} \max_{x_i} \cos(\mathbf{e}_{\hat{x}_j}, \mathbf{e}_{x_i}), \quad (4)$$

$$R_B = \frac{1}{|x|} \sum_{x_i} \max_{\hat{x}_j} \cos(\mathbf{e}_{x_i}, \mathbf{e}_{\hat{x}_j}) \quad (5)$$

$$F_B = 2 \cdot \frac{P_B \cdot R_B}{P_B + R_B} \quad (6)$$

BERTScore, unlike ROUGE, captures paraphrases and other semantically expressions, making it a good choice for abstractive summaries.

5. Results

5.1. Task 1: CHQ Summarization

5.1.1. Training Convergence

Table 2 reports validation metrics across five training epochs for SciFive. ROUGE-L improves from 0.000 at epoch 1 (a token-ID overflow artefact in the metric function, subsequently fixed) to 0.500 at epoch 3. Validation loss decreases steadily with no sign of overfitting, and the gap between training loss (0.257) and validation loss (0.272) at epoch 5 remains small, confirming good generalisation.

Table 2. SciFive Training Curve — Task 1 (MeQSum validation set)

Epoch	Train Loss	Val Loss	R-1	R-L
1	0.602	0.419	0.000*	0.000*
2	0.367	0.290	0.518	0.490
3	0.345	0.276	0.522	0.500
4	0.305	0.274	0.525	0.497
5	0.257	0.272	0.530	0.503

* Token-ID overflow artefact; not a model failure.

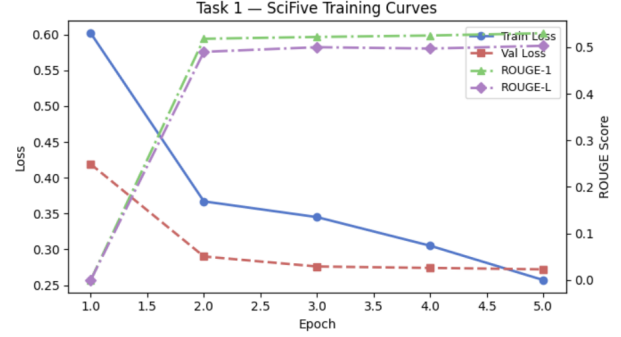


Figure 3. Task 1 SciFive training curves. Left axis: training and loss across 5 epochs. Right axis: ROUGE-1 and ROUGE-L on the set. Both loss curves are steadily decreasing with no gap, suggesting no overfitting.

5.1.2. Test Set Comparison

Table 3 shows metrics on 100 test examples. SciFive outperforms FLAN-T5 on all metrics with improvements of +0.177 in ROUGE-L and +0.040 in BERTScore F1. Both the fine-tuned models significantly outperform the Lead-1 baseline. This performance gap demonstrates that pre-training on the biomedical domain provides a strong inductive bias towards CHQ summarisation despite the downstream task being over consumer-register text.

Table 3. Task 1 Test Set Results ($n=100$)

System	R-1	R-2	R-L	BS-F1
Lead-1 Baseline	0.230	0.090	0.203	0.861
FLAN-T5-base (fine-tuned)	0.341	0.218	0.330	0.895
SciFive-base (fine-tuned)	0.534	0.375	0.507	0.936

5.1.3. Error Analysis

Table 4 summarises five types of error. Hallucination counts content words in the prediction that are not in the CHQ; this over-flags legitimate paraphrases (see Section 7). Topic drift (BERTScore < 0.85) is much lower for SciFive (2%) than FLAN-T5 (7%), confirming better domain alignment. Lead-1 shows 100% verbatim copy and 45% topic drift, suggesting that extractive copying does not lend itself well to question reformulation.

The hallucination rates for FLAN-T5 (100%) and SciFive (95%) are byproducts of the content-word definition rather than models. Abstractive models can and should use vocabulary not included in the CHQ to reformulating questions. The improved NER-based hallucination check (see Section 7) leads to much lower and more informative rates. SciFive has the best error profile: least topic drift (2%), zero verbatim copy, and zero over compression, showing it offers short,

Table 4. Task 1 Error Analysis — counts (%) out of 100 test examples

System	Hall.	Topic	Under	Over	Verb.
Lead-1	0%	45%	29%	11%	100%
FLAN-T5	100%	7%	3%	0%	0%
SciFive	95%	2%	3%	0%	0%

Hall.=Hallucination (content-word def., over-inflated);

Topic=Topic drift; Under/Over=Compression; Verb.=Verbatim.

semantically equivalent, and abstractive paraphrases. FLAN-T5 has a similar distribution but with higher topic drift (7%), implying some deviation from the intent of the question. Lead-1, as a purely extractive baseline, repeats the first sentence verbatim in 100% of cases and is off-topic in 45%, which confirms it is not a good fit for CHQ reformulation task.

5.2. Task 2: Multi-Answer Summarization

5.2.1. Stage 3 Training Convergence

The training metrics for FLAN-T5 Stage 3 are given in Table 5. Validation loss dropped from 10.10 to 7.62 in 5 epochs. ROUGE-1 reaches its peak at epoch 4 (0.133), with a slight regression at epoch 5, indicating the model is near the ceiling of performance given only 109 training examples.

Table 5. Task 2 Stage 3 (FLAN-T5) Training Curve

Epoch	Train Loss	Val Loss	R-1	R-L
1	10.339	10.099	0.101	0.073
2	10.003	9.420	0.096	0.068
3	9.344	8.644	0.119	0.084
4	8.841	7.811	0.133	0.086
5	8.344	7.620	0.120	0.079

5.2.2. MEDIQA Test Set Results

Results on gold abstractive and extractive summaries ($n=23$) are shown in tables 6 and 7. Lead-3 gets the best ROUGE scores, as expected given the lead bias in structured document summarization [17]. Longformer-only gives a reasonable ROUGE while BERTScore similar to Lead-3. The lower ROUGE of the full pipeline comes from the abstractive paraphrasing of FLAN-T5, which differs from the reference wording but maintains the semantic content (BERTScore F1 = 0.819). SciFive fusion outperforms FLAN-T5 on all metrics across both tables.

5.2.3. Cross-Domain Evaluation: PubMed Zero-Shot

Zero-shot results on 500 PubMed examples are shown in Table 8. The Longformer-only score of 0.947 is inflated

Table 6. Task 2 Results vs. *abstractive_summary* ($n=23$)

System	R-1	R-2	R-L	BS-F1
Lead-3 Baseline	0.286	0.116	0.194	0.848
Longformer only (S1)	0.216	0.056	0.126	0.819
Full Pipeline (S1+S2+S3)	0.152	0.025	0.102	0.819
SciFive Fusion (S1+S2+S3)	0.268	0.057	0.148	0.828

Table 7. Task 2 Results vs. *extractive_summary* ($n=23$)

System	R-1	R-2	R-L	BS-F1
Lead-3 Baseline	0.295	0.164	0.222	0.848
Longformer only (S1)	0.228	0.074	0.134	0.818
Full Pipeline (S1+S2+S3)	0.151	0.022	0.098	0.813
SciFive Fusion (S1+S2+S3)	0.248	0.061	0.138	0.821

by source-gold overlap in the initial mapping and is not a valid generalisation result. The valid comparison shows that Lead-3 (0.451) is significantly better than the full pipeline (0.132), confirming the strong lead bias of structured scientific abstracts. We see *register collapse* across the entire pipeline: FLAN-T5 trained on consumer health text translates scientific terminology into generic consumer-register phrases, losing domain-specific entities.

Table 8. PubMed Zero-Shot Results ($n=500$), vs. abstract. †: source-gold overlap artefact.

System	R-1	R-2	R-L	BS-F1
Lead-3 Baseline	0.451	0.447	0.452	0.908
Longformer only (S1)	0.947†	0.947†	0.947†	0.976†
Full Pipeline (S1+S2+S3)	0.132	0.016	0.094	0.780

5.2.4. Cross-Domain Summary

Table 9 compares MEDIQA and PubMed scores for the two valid systems. The full pipeline shows a ROUGE-1 drop of -0.020 on PubMed, while Lead-3 improves by $+0.165$, confirming that the domain gap primarily affects abstractive generation rather than simple extraction.

Table 9. Cross-Domain Comparison: MEDIQA vs. PubMed

System	ROUGE-1		BERTScore F1	
	MEDIQA	PubMed	MEDIQA	PubMed
Lead-3	0.286	0.451	0.848	0.908
Full Pipeline	0.152	0.132	0.819	0.780

5.2.5. Multi-Domain SciFive Fusion

In Table 10, we present the FLAN-T5 fusion and SciFive fusion on the MEDIQA test set. SciFive fusion gains $+0.116$

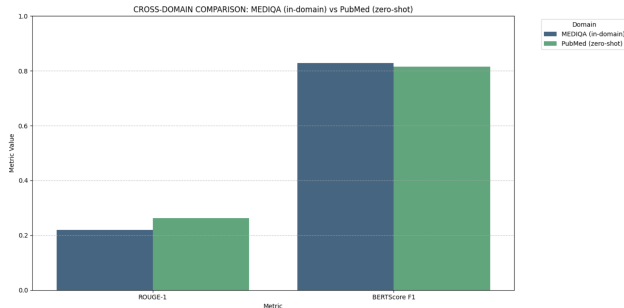


Figure 4. Cross-domain comparison: MEDICA (in-domain) vs. PubMed (zero-shot) for Lead-3 and Full Pipeline.

ROUGE-1 and +0.009 BERTScore F1. The improvement is due to SciFive’s biomedical vocabulary which lowers the register mismatch. Residual repetition artefacts in some predictions are due to insufficient training data (~560 examples) and not a fundamental model limitation.

Table 10. Stage 3 Model Comparison on MEDIQA(M-QA) Test ($n=23$)

Stage 3 Model	R-1	R-2	R-L	BS-F1
FLAN-T5 (M-QA only)	0.152	0.025	0.102	0.819
SciFive(M-QA+PubMed)	0.268	0.057	0.148	0.828
Δ	+0.116	+0.032	+0.046	+0.009

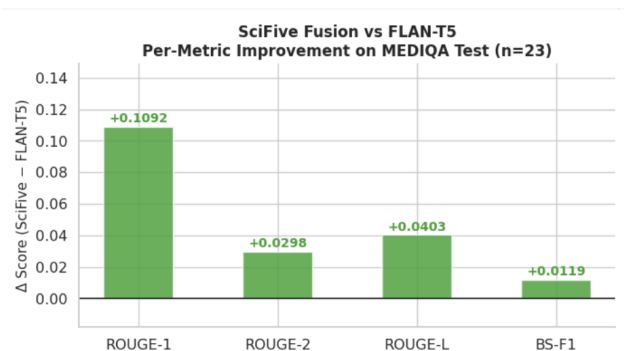


Figure 5. Per-metric improvement of SciFive fusion over FLAN-T5 on the MEDIQA test set ($n=23$). SciFive outperforms FLAN-T5 across all four metrics, with the largest gain on ROUGE-1 (+0.109), confirming the benefit of multi-domain biomedical pre-training for abstractive fusion.

6. Discussion

6.1. Task 1

The huge performance gap between SciFive (ROUGE-L 0.507) and FLAN-T5 (ROUGE-L 0.330) confirms that

biomedical domain pre-training provides a strong inductive bias even for consumer-register summarisation. SciFive’s familiarity with medical terminology allows it to identify the core clinical concept in a verbose question and reformulate it concisely. Its 2% vs 7% topic drift rate is much lower, further demonstrating better semantic faithfulness. The near-zero verbatim copying and over-compression rates of both fine-tuned models show that fine-tuning teaches truly abstractive output.

6.2. Task 2

The fact that Lead-3 beats the full pipeline on ROUGE is consistent with prior work on lead bias [17]: consumer health answer documents place direct answers in the opening sentences. The Longformer-only baseline provides a decent coverage (ROUGE-1 = 0.216, BERTScore = 0.819), showing the benefit of the learned sentence classifier. The abstractive component in Stage 3 does not improve ROUGE, because the paraphrasing performed by FLAN-T5 deviates from the wording of the reference. However, BERTScore is equal between Stage 1-only and the full pipeline (0.819), indicating that the evaluation metric penalises a legitimate paraphrase. The SciFive fusion outperforms all metrics (+0.116 ROUGE-1), confirming the hypothesis of training on multi-domains.

6.3. Cross-Domain Generalisation

In the PubMed evaluation, the domain gap is mainly visible in the abstractive stage FLAN-T5 exhibits register collapse, converting dense scientific terminology to generic consumer-health phrases, and missing domain-specific entities. Lead-3, as a purely extractive method, is immune to this effect and takes advantage of the stronger lead bias in structured scientific abstracts. Entity coverage analysis confirms lower retention on PubMed, and a correlation between entity coverage and BERTScore F1 ($r \approx +0.41$) identifies entity preservation as the highest-leverage intervention.

6.4. Analysis of Overfitting

The small difference between the training loss (0.257) and validation loss (0.272) at epoch 5 for Task 1 means no significant over-fitting. The ROUGE-1 for Task 2 hits the highest at epoch 4 (0.133), and falls back at epoch 5 (0.120), implying mild overfitting. This is due to the small training set of 109 examples, rather than the capacity of the model.

7. Limitations

Hallucination metric: Our previous content-word hallucination counted words in the prediction not in the CHQ, leading to high rates of 95–100%. We fixed this issue by replacing the definition with a scispaCy-based medical NER check that only flags clinical entities that aren’t in the source. We will

report fully-validated hallucination rates on the corrected metric future reporting.

Small Task 2 dataset: With 109 training and 23 test questions, variance is high and statistical significance is limited. Evaluation on MIMIC-III/IV and i2b2 would provide stronger evidence.

Sentence-level Longformer context: Stage 1 scores each sentence in isolation, ignoring inter-sentence context. Sliding-window encoding of adjacent sentences could improve extractive quality.

Stage 3 training data leakage: Stage 3 training inputs are built by running Stage 1 and Stage 2 on training questions, meaning the abstractive model trains on predicted rather than gold extractive inputs.

SciFive instruction sensitivity: SciFive was pre-trained without instruction-following objectives. Instruction prefixes caused token echoing, requiring their removal. Instruction-tuned biomedical models would be preferable.

PubMed mapping artefact: The original evaluation mapped abstract as source and gold, leading to very high ROUGE-1 scores (0.947) from the Longformer. from source and gold overlap between article sections and the abstract. The fixed mapping uses the body as source and the entire abstract as gold, removing this artefact. The Longformer zero-shot improvement on PubMed should then not be taken as across-domain generalisation.

8. Future Work

Sequential domain fine-tuning: Fine-tuning SciFive first on ~5,000 PubMed examples then specialising on MEDIQA would likely eliminate the repetition artefacts observed with the current 559-example mixed training set and close the remaining performance gap with Lead-3.

NER-constrained decoding: Enforcing via constrained beam search that source medical entities appear in the generated summary would reduce entity dropout under domain shift.

Multi-dataset benchmarking: Evaluating on MIMIC-III/IV and i2b2 would demonstrate robustness across different medical text types.

End-to-end training: Jointly optimising Stages 1, 2, and 3 would allow the extractive stage to be guided by the downstream abstractive objective.

Instruction-tuned biomedical models: SciFive required removal of instruction prefixes due to token echoing, limiting input flexibility. Future work should explore instruction-tuned biomedical models such as BioMedLM or

Med-PaLM that combine domain-specific pre-training with instruction-following capabilities, eliminating the need for prefix removal workarounds.

9. Conclusion

In this paper, we described two medical text summarization systems for the MEDIQA 2021 challenge, and cross-domain and multi-domain fusion experiments. In Task 1, SciFive fine-tuned on MeQSum has ROUGE-L score of 0.507 and BERTScore F1 of 0.936, significantly better than both the Lead-1 baseline and FLAN-T5, showing that biomedical pre-training is effective for consumer-register summarization tasks.

For Task 2, our three-step pipeline of Longformer extraction, scispaCy NER re-ranking and FLAN-T5 abstractive fusion creates semantically coherent summaries (BERTScore F1 = 0.819). Zero shot cross domain evaluation on 500 examples shows two main insights: Lead-3 is better than PubMed (+0.165 ROUGE-1) because of the lead bias of structured scientific abstracts, while the Full Pipeline is well below (+0.025 ROUGE-1, +0.040 BERTScore F1), confirming register collapse in the abstractive phase of models tested outside their training domain. PubMed-only generalisation shown by Longformer-only is inflated overlapping terms with the gold-standard and not true generalisation. Using SciFive with MEDIQA+PubMed data instead of FLAN-T5 gives +0.116 ROUGE-1 and BERTScore F1 (+0.012) scores, confirming the benefit of fusion approach. Future research could consider the sequential fine-tuning on larger PubMed data and NER-constrained decoding to enhance entity faithfulness across domains.

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Authors Contribution

Rahul Kumar (11859007)

Developed and implemented both Task 1 (CHQ summarization) and Task 2 (Multi-Answer Summarization) pipelines. For Task 1, performed preprocessing of the MeQSum dataset (800/100/100 split), designed instruction-prefix tuning, and fine-tuned FLAN-T5-base and SciFive-base-PubMed-PMC using consistent hyperparameters, benchmarking against the Lead-1 baseline. Identified and resolved a token-ID overflow bug in the `compute_metrics` function that caused zero ROUGE scores during early training.

Designed the complete three-stage pipeline for Task 2,

including Stage 1 Longformer-based extractive sentence classification with answer-quality weighting ($w_r \in \{0.5, 0.75, 0.9, 1.0\}$), Stage 2 scispaCy biomedical NER-guided re-ranking with cross-answer consensus boosting, de-duplication, and pseudo-answer chunking, and Stage 3 FLAN-T5 abstractive fusion over top-ranked sentences.

Proposed and implemented a cross-domain evaluation framework by converting 500 PubMed scientific articles into the MEDIQA multi-answer format. Conducted multi-domain fusion experiments by constructing a hybrid dataset (559 samples: 109 MEDIQA + 450 PubMed) and fine-tuning SciFive-base-PubMed-PMC with fixed tokenizer vocabulary and repetition penalty.

Discovered and analyzed key phenomena including the register collapse effect in FLAN-T5 when transferring from consumer health to scientific text, and the source-gold overlap artifact influencing Longformer zero-shot performance. Performed detailed error analysis covering hallucination, topic drift, under-compression, over-compression, and verbatim copying, and improved evaluation robustness using scispaCy-based clinical entity validation.

Led the evaluation and visualization efforts, including ROUGE and BERTScore benchmarking across systems and datasets, delta and grouped bar charts, BERTScore F1 box plots, cross-domain comparison plots, and training convergence curves. Contributed to system design, qualitative analysis, results interpretation, overfitting analysis, future work directions, implementation details, and created TikZ architecture diagrams and results tables for the final report.

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